Day3 演習2

```
import matplotlib.pyplot as plt
import numpy as np
from scipy.integrate import solve_ivp

def selkov_model( t , init , *param ):
    x = init[ 0 ]
    y = init[ 1 ]

    [ a , b ] = param


    v1 = b
    v2 = a * y + (x ** 2) * y
    v3 = x

    dxdt = v2 - v3
    dydt = v1 - v2

    return [ dxdt , dydt ]


init = [ 1.0 , 1.0 ]
t_span = [ 0 , 100 ]
t_eval = np.linspace( t_span[0] , t_span[1] , 10000 )

a = 0.06
b = 0.6

param = [ a , b ]

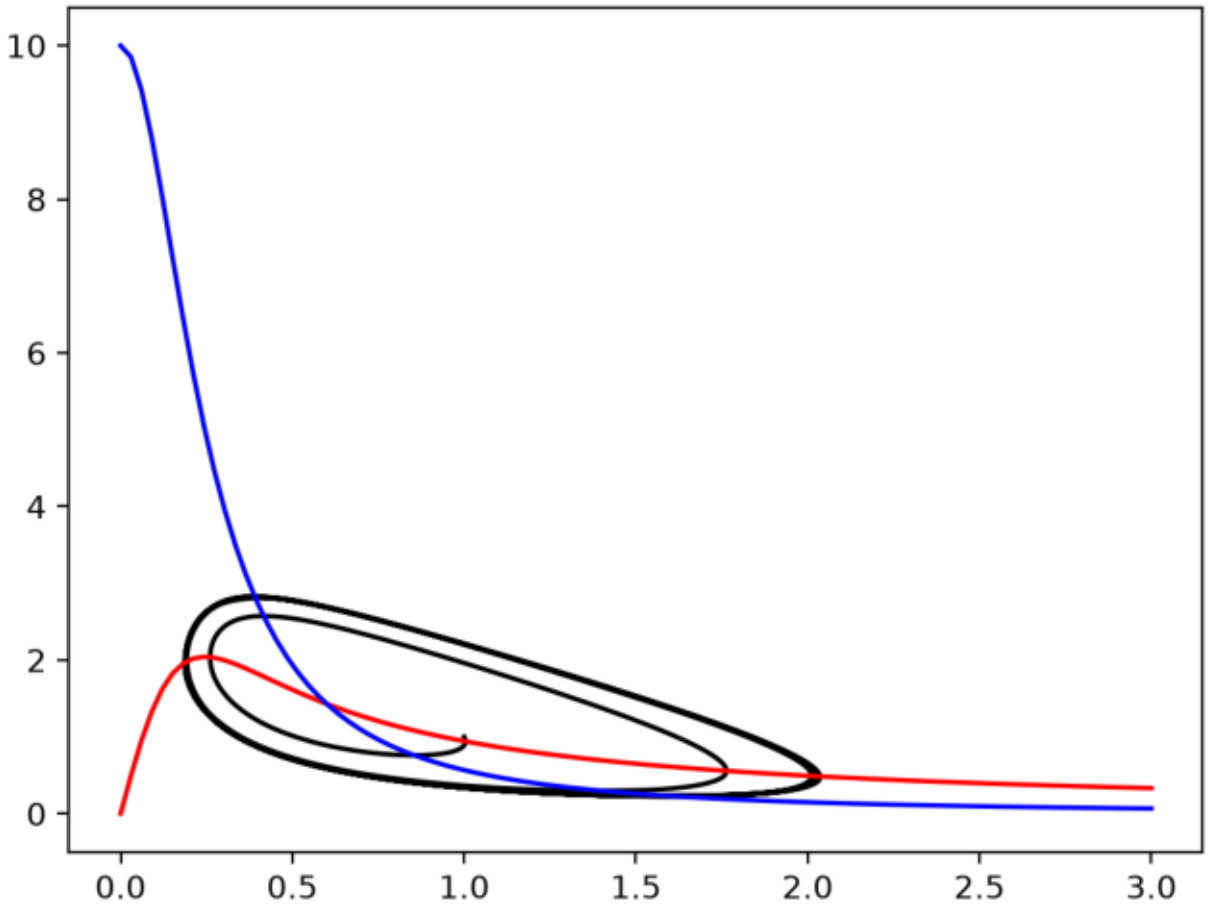
solution = solve_ivp( selkov_model, t_span, init, t_eval=t_eval, method='RK45',
                      args=param )

ADP = solution.y[0]
F6P = solution.y[1]

plt.plot(ADP, F6P, 'k-')

x = np.linspace( 0, 3, 100 )
x_null = x / ( a + x ** 2 ) # The nullcline for dydt
y_null = b / ( a + x ** 2) # The nullcline for dydt

plt.plot( x, x_null, 'r-')
plt.plot( x, y_null, 'b-')
plt.show()
```


10
8
6
4
2
0
0.0
0.5
1.0
1.5
2.0
2.5
3.0